# Statistical Data Analysis

## Lecture 9: Nonparametric methods part II

**Marc Corstanje**

Periods 4 & 5

# Where Are We?

- Chapter 6: Nonparametric methods
  - Section 6.1 – The one-sample problem
  - Section 6.2 – Asymptotic efficiency
  - Section 6.3 – Two-sample problems
  - Section 6.4 – Tests for correlation
- Chapter 7: Analysis of categorical data

# Asymptotic Relative Efficiency

# Recall – Asymptotic Efficiency

- Statistical problem for $X_1, \ldots, X_n \overset{i.i.d.}{\sim} F$
  - Test $H_0 : F \in \mathcal{F}_0$ (e.g. class of all distributions with median $m_0$)
  - Reformulate to location shifts $H_0 : \theta \leq 0$ vs. $H_1 : \theta > 0$
  - Power $\pi(\theta) = \pi(F_\theta) = \mathbb{P}_\theta(H_0 \text{ is rejected})$
- For an asymptotically normal test statistic $T_n$
  - Asymptotic mean, standard deviation $\mu(\theta)$, $\sigma(\theta)$.
  - $\sqrt{n}\frac{T_n - \mu(\theta)}{\sigma(\theta)} \rightsquigarrow N(0, 1)$.
  - If $\theta_n = h/\sqrt{n}$

$$\text{power} \overset{\text{large } n}{\approx} 1 - \Phi\left(\xi_{1-\alpha} - \frac{\mu'(0)}{\sigma(0)} \cdot h\right)$$

  - $\mu'(0)/\sigma(0)$ is also called the slope of the test
  - When $h$ increases, the power increases faster when slope is larger
  - A smaller test slope can be compensated by a larger sample size

# Asymptotic Relative Efficiency – Idea

- Consider two tests based on the test statistic $T_n$ resp. $\tilde{T}_n$
  - Asymptotic mean and standard deviation: $\mu(\cdot)$ resp. $\tilde{\mu}(\cdot)$, $\sigma(\cdot)$ resp. $\tilde{\sigma}(\cdot)$
- Asymptotic relative efficiency: square of ratio of test slopes

$$\mathrm{are}(T_n, \tilde{T}_n) = \left( \frac{\mu'(0)/\sigma(0)}{\tilde{\mu}'(0)/\tilde{\sigma}(0)} \right)^2$$

  - If $\mathrm{are}(T_n, \tilde{T}_n) > 1$, then $T_n$ is more efficient than $\tilde{T}_n$
  - If $\mathrm{are}(T_n, \tilde{T}_n) < 1$, then $T_n$ is less efficient than $\tilde{T}_n$

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  - If $\text{are}(T_n, \tilde{T}_n) > 1$, then $T_n$ is more efficient than $\tilde{T}_n$
  - If $\text{are}(T_n, \tilde{T}_n) < 1$, then $T_n$ is less efficient than $\tilde{T}_n$
- $\text{are}(T_n, \tilde{T}_n)$ depends on the type of shift alternatives $F_\theta(\cdot) = F_0(\cdot - \theta)$
  - As the asymptotic distribution of $T_n$ and $\tilde{T}_n$ for $\theta = 0$ depends on $F_0$

| | ***t*** | ***s*** | ***w*** |
|---|---|---|---|
| ***t*** | 1 | | |
| ***s*** | $2/\pi$ | 1 | |
| ***w*** | $3/\pi$ | $3/2$ | 1 |

**$N(0,1)$**

| | ***t*** | ***s*** | ***w*** |
|---|---|---|---|
| ***t*** | 1 | | |
| ***s*** | $\pi^2/12$ | 1 | |
| ***w*** | $\pi^2/9$ | $4/3$ | 1 |

**Logistic**

| | ***t*** | ***s*** | ***w*** |
|---|---|---|---|
| ***t*** | 1 | | |
| ***s*** | $1/3$ | 1 | |
| ***w*** | 1 | 3 | 1 |

**Uniform**

| | ***t*** | ***s*** | ***w*** |
|---|---|---|---|
| ***t*** | 1 | | |
| ***s*** | 2 | 1 | |
| ***w*** | $3/2$ | $3/4$ | 1 |

**Laplace**

AREs (row over column) of the $t$ (***t***), sign (***s***) and Wilcoxon (***w***) test for shift alternatives of different $F_0$

# Asymptotic Relative Efficiency – Simulation Experiments

```r
### Power simulation function
# We simulate normal data under the shift alternative (using theta > 0)
are_sim <- function(B, n, m_0 = 0, theta, sig = 1){
  p_t <- p_s <- p_w <- numeric(B)
  for(i in 1:B){
    x <- rnorm(n, mean = m_0 + theta, sd = sig)
    # The true median is m = m_0 + theta (m > m_0, as theta > 0)
    # H_0: m <= m_0, H_1: m > m_0
    p_t[i] <- t.test(x, alternative = "g", mu = m_0)$p.value
    p_s[i] <- binom.test(sum(x > m_0), n, p = 0.5, alternative = "g")$p.value
    p_w[i] <- wilcox.test(x, alternative = "g", mu = m_0)$p.value
  }
  # Power approximated via fraction of (correctly) rejected tests
  power_t <- mean(p_t < 0.05)
  power_s <- mean(p_s < 0.05)
  power_w <- mean(p_w < 0.05)
  # AREs (ratios)
  c(are_st = power_s/power_t, are_wt = power_w/power_t)
}

### AREs computed via simulation
set.seed(123)
are_sim(B = 10^3, n = 10^2, m_0 = 0, theta = 0.1, sig = 1)
```

```
##    are_st    are_wt
## 0.6549020 0.9843137
```

```r
### Theoretical AREs
c(2/pi, 3/pi)
```

```
## [1] 0.6366198 0.9549297
```

# Asymptotic Relative Efficiency – Further Considerations

- ARE can be interpreted in terms of sample sizes: ratio of the numbers of observations needed for the two tests to have approximately equal asymptotic power (given $\alpha$)
  - Take $T_n$ and $\tilde{T}_m$ with $\frac{m}{n} = \text{are}(T_n, \tilde{T}_m) \rightsquigarrow$ the tests have ca. equal power

```
B <- 10^3; p_t <- p_s <- p_sm <- numeric(B)
n <- 10^2; m <- round((pi/2) * n); set.seed(123)
for(i in 1:B){
  x_n <- rnorm(n, mean = 0.1, sd = 1); x_m <- rnorm(m, mean = 0.1, sd = 1)
  p_t[i] <- t.test(x_n, alternative = "g", mu = 0)$p.value
  p_s[i] <- binom.test(sum(x_n > 0), n, p = 0.5, alternative = "g")$p.value
  p_sm[i] <- binom.test(sum(x_m > 0), m, p = 0.5, alternative = "g")$p.value
}
c(pow_t = mean(p_t < 0.05), pow_s = mean(p_s < 0.05), pow_sm = mean(p_sm < 0.05))
```

```
##  pow_t  pow_s pow_sm
##  0.277  0.173  0.244
```

# Asymptotic Relative Efficiency – Further Considerations

| | $t$ | $s$ | $w$ |
|---|---|---|---|
| $t$ | 1 | | |
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$N(0,1)$

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| $t$ | 1 | | |
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Logistic

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Uniform

| | $t$ | $s$ | $w$ |
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| $t$ | 1 | | |
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Laplace

- The Wilcoxon signed rank test is a serious competitor of the $t$-test
  - Small loss of efficiency when $t$ is optimal: $\text{are}(\boldsymbol{w}, \boldsymbol{s}) = 3/\pi \approx 0.95$
  - Potential big gain in efficiency when $t$ is not optimal (e.g. Laplace data)
- The sign test is nonparametric against all alternatives
  - Wilcoxon and $t$ require symmetry resp. normality

# Asymptotic Relative Efficiency – Further Considerations

- ARE can be interpreted in terms of sample sizes: ratio of the numbers of observations needed for the two tests to have approximately equal asymptotic power (given $\alpha$)
  - Take $T_n$ and $\tilde{T}_m$ with $\frac{m}{n} = \text{are}(T_n, \tilde{T}_m) \rightsquigarrow$ the tests have ca. equal power

```
B <- 10^3; p_t <- p_s <- p_sm <- numeric(B)
n <- 10^2; m <- round((pi/2) * n); set.seed(123)
for(i in 1:B){
  x_n <- rnorm(n, mean = 0.1, sd = 1); x_m <- rnorm(m, mean = 0.1, sd = 1)
  p_t[i] <- t.test(x_n, alternative = "g", mu = 0)$p.value
  p_s[i] <- binom.test(sum(x_n > 0), n, p = 0.5, alternative = "g")$p.value
  p_sm[i] <- binom.test(sum(x_m > 0), m, p = 0.5, alternative = "g")$p.value
}
c(pow_t = mean(p_t < 0.05), pow_s = mean(p_s < 0.05), pow_sm = mean(p_sm < 0.05))
```

```
##  pow_t  pow_s pow_sm
##  0.277  0.173  0.244
```

# Two-Sample Problems

# Two-Sample Problems – Idea

- Goal: compare two univariate samples with each other
  - Paired data: $(X_1, Y_1), \ldots (X_n, Y_n)$
  - Unpaired data: $X_1, \ldots, X_m$ and $Y_1, \ldots, Y_n$

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  - Paired data: $(X_1, Y_1), \ldots (X_n, Y_n)$
  - Unpaired data: $X_1, \ldots, X_m$ and $Y_1, \ldots, Y_n$
- Unpaired: two independent samples $X_1, \ldots, X_m \overset{i.i.d}{\sim} F$ and $Y_1, \ldots, Y_n \overset{i.i.d}{\sim} G \rightsquigarrow$ Test hypotheses $H_0 : F = G$ vs. $H_0 : F \neq G$
  - One-sided alternatives are also possible (*stochastically* smaller/larger)
  - Assume $F$ and $G$ are continuous (if ties are present, use pseudo ranks)
  - To test shift-alternatives: median test, Wilcoxon two-sample test
  - To test different 'shapes': Kolmogorov-Smirnov two-sample test
  - Permutation test(s)

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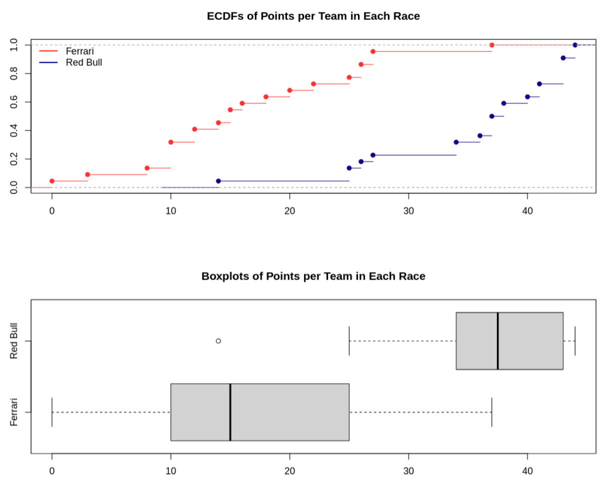
  - To test shift-alternatives: median test, Wilcoxon two-sample test
  - To test different 'shapes': Kolmogorov-Smirnov two-sample test
  - Permutation test(s)
- Paired: $n$ independent bivariate random vectors $(X_1, Y_1), \ldots, (X_n, Y_n)$ $\rightsquigarrow$ Test hypotheses $H_0$ : independence vs. $H_1$ : dependence
  - $H_0 : X_i$ and $Y_i$ are independent, $i = 1, \ldots, n$
  - Spearman and Kendall rank correlation tests
  - Permutation test(s)

# (Unpaired) Two-Sample Problem – Example: F1 in 2023

- Is Ferrari's performance significantly worse than Red Bull's?
  - $m = n = 22$ (races), sum points of both drivers in each race
  - Formally: $H_1$ : $\texttt{ferrari} <_{st} \texttt{redbull}$, i.e. $F_F(u) > F_{RB}(u)$ for all $u$



# Median Test

- Test statistic: $Z = \#(i = 1, \ldots, m : X_i \leq \text{med}(X_1, \ldots, X_m, Y_1, \ldots, Y_n))$
- Null distribution: $Z \sim \text{Hypergeom}(m+n, m, p)$ with $p = \lfloor \frac{m+n+1}{2} \rfloor$ draws without replacement
  - Analogous distribution under $H_0$ as: randomly select (without replacement) $p$ elements from the combined sample and count the number $Z$ of those belonging to the '$X$ sample'
  - Large values of $Z$ are more likely when $\mathbb{P}(Y > X) > 1/2$
  - Nonparametric test
- Rarely used nowadays
  - Has smaller power than Wilcoxon two-sample (Mann-Whitney) test
  - Not readily available in R
  - However, it can easily be coded using the cumulative distribution function of the hypergeometric distribution (see `?"phyper"`) to compute $p$-values

# Wilcoxon Two-Sample Test

- Test statistic: $W = \sum_{i=1}^{n} R_i$, where $R_i$ denotes the rank of $Y_i$ in the combined sample. Equivalent test statistics are:
  - $U = \sum_{i=1}^{m} \sum_{j=1}^{n} 1_{\{X_i < Y_j\}} = W - \frac{1}{2} n(n+1)$
  - $\tilde{W} = \sum_{i=1}^{m} S_i$, where $S_i$ denotes the rank of $X_i$ in the combined sample
  - $\tilde{U} = \tilde{W} - \frac{1}{2} m(m+1) \rightsquigarrow$ used by R

# Wilcoxon Two-Sample Test

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  - $U = \sum_{i=1}^{m} \sum_{j=1}^{n} 1_{\{X_i < Y_j\}} = W - \frac{1}{2}n(n+1)$
  - $\tilde{W} = \sum_{i=1}^{m} S_i$, where $S_i$ denotes the rank of $X_i$ in the combined sample
  - $\tilde{U} = \tilde{W} - \frac{1}{2}m(m+1) \rightsquigarrow$ used by R
- Null distribution: normal approximation can be used

```
# Note direction of the alternative: "less" (!)
# This is as R uses what we called tilde(U)) (called W in R output)
# Test statistic tilde(U) depends on sum of ranks of X sample
wilcox.test(ferrari, red_bull, alternative = "less", paired = FALSE)
```

```
## Warning in wilcox.test.default(ferrari, red_bull, alternative = "less", :
## cannot compute exact p-value with ties

##
##  Wilcoxon rank sum test with continuity correction
##
## data:  ferrari and red_bull
## W = 34, p-value = 5.331e-07
## alternative hypothesis: true location shift is less than 0
```

# Kolmogorov-Smirnov Two-Sample Test

- Test statistic:

$$D = \sup_{-\infty < x < \infty} \left| \hat{F}_m(x) - \hat{G}_n(x) \right|$$
$$= \max_{1 \le i \le n} \max \left\{ \left| \frac{1}{m}(R_{(i)} - i) - \frac{i}{n} \right|, \left| \frac{1}{m}(R_{(i)} - i) - \frac{i-1}{n} \right| \right\}$$

  - $R_{(i)}$ denotes the rank of $Y_i$ in the combined sample

# Kolmogorov-Smirnov Two-Sample Test

- Test statistic:

$$D = \sup_{-\infty < x < \infty} \left| \hat{F}_m(x) - \hat{G}_n(x) \right|$$
$$= \max_{1 \le i \le n} \max \left\{ \left| \frac{1}{m}(R_{(i)} - i) - \frac{i}{n} \right|, \left| \frac{1}{m}(R_{(i)} - i) - \frac{i-1}{n} \right| \right\}$$

  - $R_{(i)}$ denotes the rank of $Y_i$ in the combined sample
- Null distribution: depends on $m$ and $n$, but not on $F$ or $G$
  - We reject $H_0 : F = G$ for large values of $D$
  - See below for one-sided alternative in R

```
# Note direction of the alternative: "greater" (!)
ks.test(ferrari, red_bull, alternative = "greater")

##
##  Exact two-sample Kolmogorov-Smirnov test
##
## data:  ferrari and red_bull
## D^+ = 0.72727, p-value = 1.846e-06
## alternative hypothesis: the CDF of x lies above that of y
```

# Permutation Test – Permutations

- Denote by $Z_{(1)} \leq \cdots \leq Z_{(m+n)}$ the ordered combined sample and by $Z_{(\pi_1)}, \ldots, Z_{(\pi_{m+n})}$ a given permutation thereof
  - That is, $(\pi_1, \ldots, \pi_{m+n})$ is a permutation of $\{1, 2, \ldots, m+n\}$.
- Then

$$P\left((X_1, \ldots, X_m, Y_1, \ldots, Y_n) = (Z_{(\pi_1)}, \ldots, Z_{(\pi_{m+1})})\right) = \frac{1}{(m+n)!}$$

  - The probability that the ordering $X_1, \ldots, X_m, Y_1, \ldots, Y_n$ is obtained by rearranging $Z_{(1)} \leq \cdots \leq Z_{(m+n)}$ by the permutation $\pi_1, \ldots, \pi_{m+n}$ equals $\frac{1}{\#\text{ permutations of } \{1,\ldots,m+n\}}$.

# Permutation test – Null distribution

- The test uses some test statistic $T$ that quantifies the difference between the two samples
  - Assume very large (or very small) values of $T$ lead to the rejection of $H_0$
  - For example, $T = \bar{X} - \bar{Y}$
  - $t$ is the observed value of $T$ in the original sample
- Null distribution: the right $p$-value (the left one is analogous) can be computed as

$$P\left(T \geq t | Z_{(1)} \leq \cdots \leq Z_{(m+n)}\right) \\ = \frac{\#\left(\text{permutations } \pi \text{ with } T\left(Z_{(\pi_1)}, \ldots, Z_{(\pi_{m+n})}\right) \geq t\right)}{(m+n)!}$$

  - There are $(m+n)!$ possible permutations $\rightsquigarrow$ can be too large
  - In practice, a subset of randomly chosen ones is then used to approximate the $p$-value $\rightsquigarrow$ bootstrap ☺

# Tests for Correlation

# Tests for Correlation – Idea

- Setup: $n$ independent bivariate random vectors $(X_1, Y_1), \ldots (X_n, Y_n)$
  $\rightsquigarrow$ Test hypotheses $H_0$ : independence vs. $H_1$ : dependence
  - Independence hypothesis: $X_i$ is independent of $Y_i$, $i = 1, \ldots, n$
  - If $(X_i, Y_i) \sim N_2(\mathbf{0}, \Sigma_{2\times 2})$, one can use $r_{xy} \overset{H_0}{\sim} t_{n-2}$

# Tests for Correlation – Idea

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  $\rightsquigarrow$ Test hypotheses $H_0$ : independence vs. $H_1$ : dependence
  - Independence hypothesis: $X_i$ is independent of $Y_i$, $i = 1, \dots, n$
  - If $(X_i, Y_i) \sim N_2(\mathbf{0}, \Sigma_{2\times 2})$, one can use $r_{xy} \overset{H_0}{\sim} t_{n-2}$
- If the normality assumption is not fulfilled, use ranks...
  - $S_1, \dots, S_n$: ranks of $X_1, \dots, X_n$ in $X_{(1)} \leq \dots \leq X_{(n)}$
  - $R_1, \dots, R_n$: ranks of $Y_1, \dots, Y_n$ in $Y_{(1)} \leq \dots \leq Y_{(n)}$
  - If $X_1, \dots, X_n$ are independent from $Y_1, \dots, Y_n$, then $S_1, \dots, S_n$ and $R_1, \dots, R_n$ are as well ($\rightsquigarrow$ random permutations of $1, \dots, n$)
  - Otherwise, we expect the two sequences of ranks to run in parallel (positive dependence) or in opposite directions (negative dependence)
  - Tests based solely on the vectors of ranks are nonparametric under $H_0$ (if ties occur, this holds conditionally on the pattern of ties)
- ... or permutations

# Spearman's Rank Correlation Test

- Test statistic: $r_s = \frac{\sum_{i=1}^{n}(R_i - \bar{R})(S_i - \bar{S})}{\left[\sum_{i=1}^{n}(R_i - \bar{R})^2 \sum_{i=1}^{n}(S_i - \bar{S})^2\right]^{1/2}}$
  - Sample correlation coefficient of the vectors of ranks
- Equivalent test statistic: $L = \sum_{i=1}^{n}(R_i - S_i)^2$
  - Null distribution of $L$: normal approximation can be used

```r
TV <- openintro::gradestv$tv
grades <- openintro::gradestv$grades
cor.test(TV, grades, method = "spearman")
```

```
## Warning in cor.test.default(TV, grades, method = "spearman"): Cannot compute
## exact p-value with ties

##
##  Spearman's rank correlation rho
##
## data:  TV and grades
## S = 4090.8, p-value = 0.002732
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##        rho
## -0.5733909
```

# Kendall's Rank Correlation Test

- Test statistic: $\tau = \frac{\sum\sum_{i \neq j} \text{sgn}(R_i - R_j)\text{sgn}(S_i - S_j)}{n(n-1)} = \frac{4N_\tau}{n(n-1)} - 1$
- Equivalent test statistic: $N_\tau = \#\{\text{pairs } i < j\text{: } [X_i < X_j \ \& \ Y_i < Y_j] \text{ or } [X_i > X_j \ \& \ Y_i > Y_j]\}$

```r
cor.test(TV, grades, method = "kendall")
```

```
## Warning in cor.test.default(TV, grades, method = "kendall"): Cannot compute
## exact p-value with ties

##
##  Kendall's rank correlation tau
##
## data:  TV and grades
## z = -3.199, p-value = 0.001379
## alternative hypothesis: true tau is not equal to 0
## sample estimates:
##       tau
## -0.469793
```

# Permutation Test for Independence

- Consider the ordered sample $Y_{(1)} \leq \cdots \leq Y_{(n)}$ and a given permutation thereof, denoted by $Y_{(\pi_1)}, \ldots, Y_{(\pi_n)}$
- The test uses some test statistic $T$ that quantifies dependence between the two samples
  - Assume large values of $T$ express positive dependence
  - For example, $T = r_{xy}$ or $T = r_s$
- Null distribution: the right $p$-value (the left one is analogous) can be computed as

$$P_{H_0}\left(T \geq t | X_1, \ldots, X_n, Y_{(1)}, \ldots, Y_{(n)}\right)$$
$$= \frac{\#\left(\text{permutations } \pi \text{ with } T\left(X_1, \ldots, X_n, Y_{(\pi_1)}, \ldots, Y_{(\pi_n)}\right) \geq t\right)}{n!}$$

  - There are $n!$ possible permutations $\rightsquigarrow$ can be too large
  - In practice, a subset of randomly chosen ones is then used to approximate the $p$-value $\rightsquigarrow$ bootstrap ☺

# Summary and wrap-up

# Summary

- Asymptotic relative efficiency
  - Compare the slope of two different tests for the same hypothesis
  - Interpretation: $\tilde{T}_n$ requires $\text{are}(T_, \tilde{T}_n) \times n$ data to be as efficient as $T_n$.
- Two-sample problems – unpaired data
  - Median test: Check whether the expected amount of $X$-data falls below the median of the pooled sample
  - Wilcoxon test: Check whether the ranks of the $Y$ data in the pooled sample are large.
  - Kolmogorov-Smirnov test: Compare ECDFs
  - Permutation test: If the data originates from the same distribution, permuting the pooled sample should not change the distribution of the test statistic
- Two-sample problems – paired data
  - Spearman's rank correlation test: Are the ranks of the $X$ and $Y$ data correlated?
  - Kendall's rank correlation test: counts concordant/disordant pairs.
  - Permutation test for independence.

# What now?

- To do for now
  - Continue with assignment 5
  - Deadline: 14 April 23:59
  - Good luck!
- Tutorials
  - Only 1 TA present in 4B05, 09:00 – 11:00.